



Chair for Data Science and
Natural Language Processing

Hybrid AI Studio @ NLP Lab

Prof. Dr. Siegfried Handschuh
Götz-Henrik Wiegand & Lorena Raichle

Agenda

Week I, 15. 09.2026

- 01 TA-Team & NLP Lab
- 02 What is the *Hybrid AI Studio* ?
- 03 Hackathon Planning & Teams
- 04 FrameX Intro
- 05 FrameX First Steps
- 06 Tasks

Teaching Assistant Team



Götz-Henrik Wiegand

- M. Sc. Computer Science
- PhD Student @ NLP Lab
- LLM Architectures



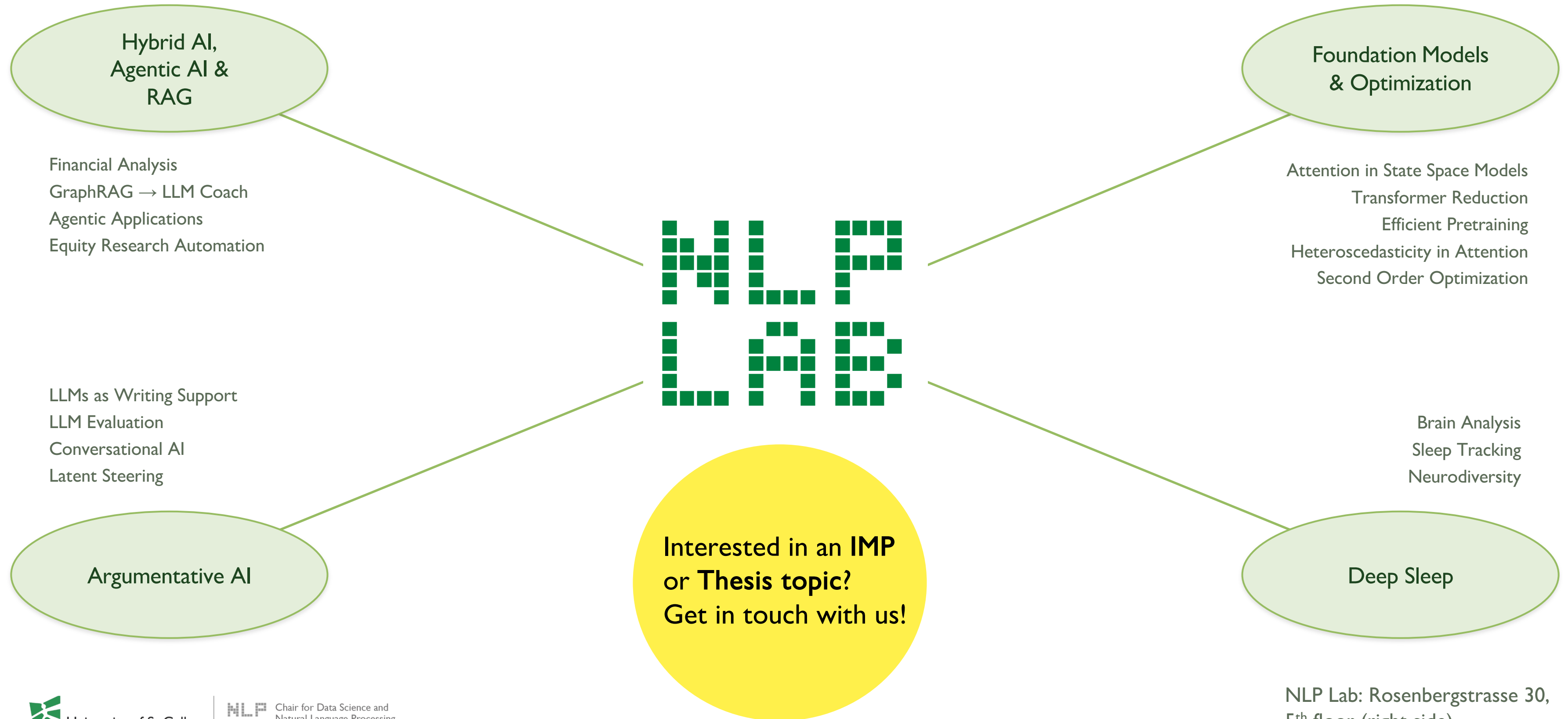
Lorena Raichle

- M. Sc. Computer Science
- PhD Student @ NLP Lab
- Token-Weighting Strategies in LLM Training

Contact the
Teaching Team
via E-Mail!

hybrid-ai@mailman.nlp-lab.ai

NLP Lab: Research Topics



NLP Lab: Quantitative Data Science

P

A Convexity-dependent Two-Phase Training Algorithm for Deep Neural Networks

P

Integrating the Attention Mechanism into State Space Models

P

Reducing the Transformer Architecture to a Minimum

Interested in an IMP
or Thesis topic?
Get in touch with us!



Research - 18.11.2025 - 10:00

HSG researchers develop more efficient method for training AI



More NLP Lab Research →

Hybrid AI Studio

Hybrid AI Studio

- Hands-on deep dive of covered lecture content
- Working in groups, Brainstorming, discussing material
- Over 3 Hackathons (within 12 semester weeks), we build an evolving system
- Studio Sessions serve as **Hackathon preparation** or **Post-Discussion**

Regular Session

Lecture

Studio

Hackathon Session

One block · 4 lectures continuous

3

Hackathons

Hybrid AI Studio to Prep & Review

Hackathon Planning

Hackathon Teams

- Teams of **3 Students**, same Teams for all 3 Hackathons!
- 3 roles per Hackathon-Team:
 - DEVELOPER
 - DOMAIN EXPERT
 - AUDITOR

→ Rotating roles after each Hackathon
- Building one **evolving system**, one system version per Hackathon

Week 03	29.09.2026	Hackathon 1	Derive from Knowledge
Week 07	10.11.2026	Hackathon 2	Connect Language and Knowledge
Week 10	01.12.2026	Hackathon 3	Plan and Adapt

3

Hackathons in 12 weeks

Working on an evolving system

Grading

90 %

Portfolio. Every Hackathon counts 30 %.

10 %

Active participation
(including Studio Sessions!)
No exam.

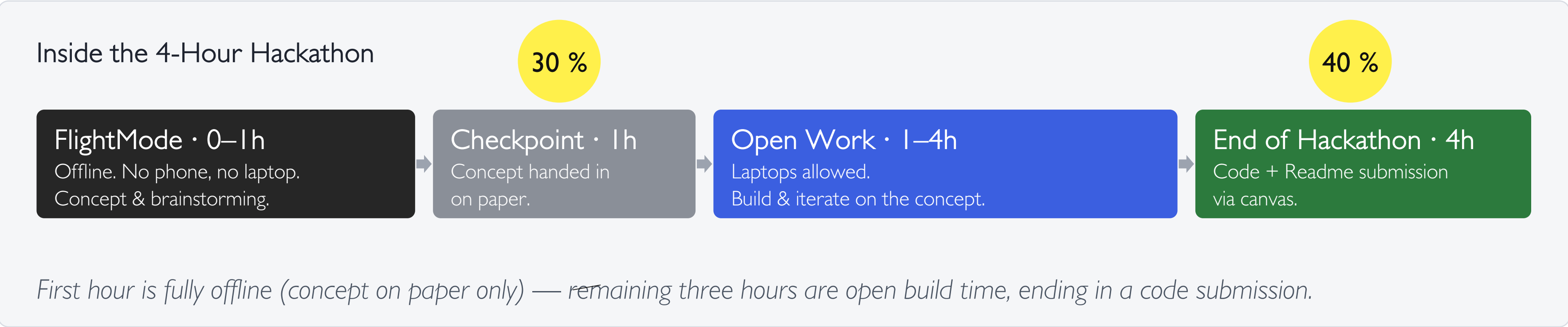
3 Deliverables per Hackathon:

30/40/30

FlightMode Docs / Code / Demo & Report

Hackathon Meta-Concept — One Cycle Across Two Weeks

Prep before · 4h hackathon in flight mode · presentation & reflection after



After the hackathon:

teams reflect and write a short report at home before the next studio session, where they demo, pitch, and submit their final report.

30 %

Grading

3 Deliverables per Hackathon:

30/40/30

FlightMode Docs / Code / Demo & Report

FlightMode Docs

- No phones, laptops or other devices!
- Problem understanding, brainstorming and task discussion
- Working on first solution ideas on paper

→ Deliverable: handing in FlightMode Docs after FlightMode phase!

Code

- Code Submission latest at 18:15 after each Hackathon
- Code must be reproducible, readable and cleaned up
- Readme must contain all necessary instructions and project overview

→ Deliverable: zip file with Code & Readme via Canvas!

Demo & Report

- Report is written between finished Hackathon and next Studio Session
- Report written in Latex with [IMRaD](#) structure (max. 8 pages content)
- Demo pitched with Slides in next Studio Session

→ Deliverable: Demo Slides & Report, both PDF!

Hackathon Teams

Hackathon Teams

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Week 10	01.12.2026	Hackathon 3	Plan and Adapt

FrameX

FrameX-Workbench

WorkspaceAccount

lorena.raichle@unisg.ch · studentSign out

Your lab

running

Stop workspace

FILES

+ New

classes-inheritance.fx

family-test.fx

joins.fx

numeric.fx

property-identity.fx

quick-starttest.fx

relations.fx

schemas.fx

tutorial.fx

Hide files

classes-inheritance.fx

Saved

Save

Run program

1 world open.

2 Person { label: String }.

3 Man extends Person.

4 Researcher extends Person.

5 isaac:Man[label -> "Isaac"].

6 isaac:Researcher.

7 ?- isaac:Person.

8 ?- isaac:Researcher.

9 expect true: isaac:Person.

10

▼ Result

completed

Compact

true

true

PASS classes-inheritance.fx:9:1 expected true, got true

1 passed, 0 failed

Executed: classes-inheritance.fx · saved version at run time

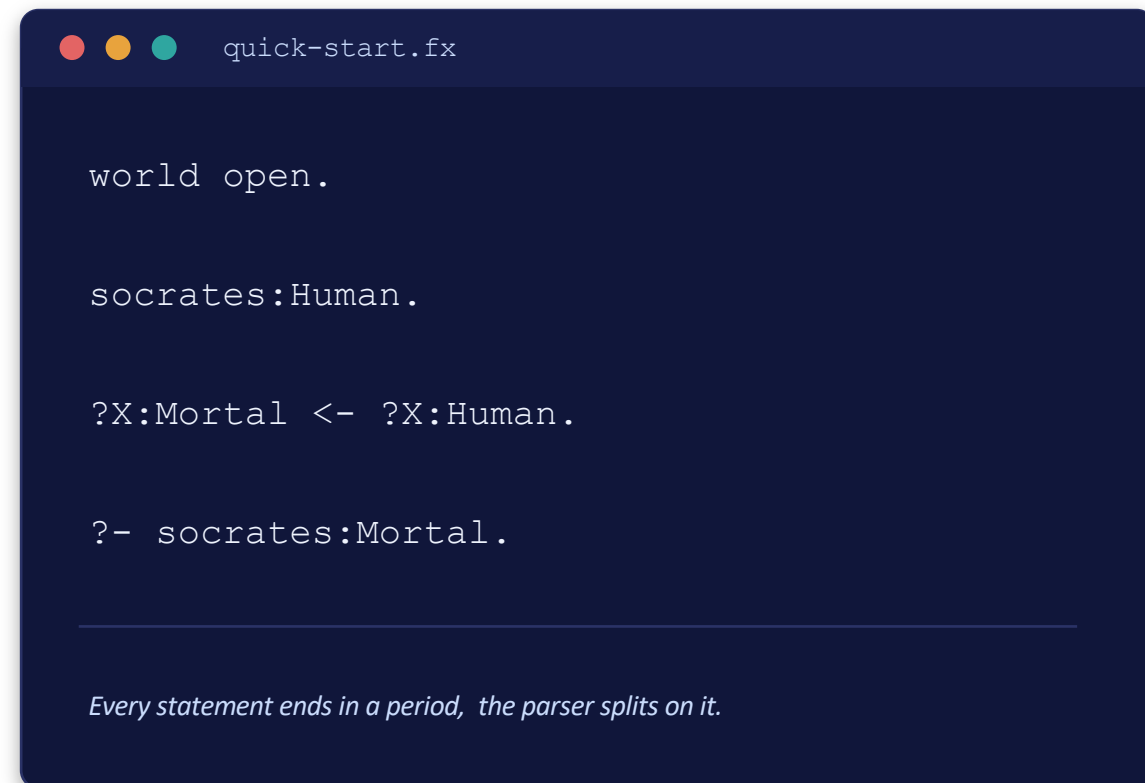
Logical truth (true, false, unknown) is separate from execution status.

► Execution details

FrameX Notation

FrameX-Workbench = editor UI + FrameX-Engine

- Every FrameX program is a short sequence of statements, each one ending in a period.



```
quick-start.fx

world open.

socrates:Human.

?X:Mortal <- ?X:Human.

?- socrates:Mortal.
```

Every statement ends in a period, the parser splits on it.



Statements end with a period

Each line above is one statement, closed by '.' (whitespace and line breaks don't matter).



A leading '?' marks a variable

?X stands for any individual. It is re-bound each time the rule is checked against a fact.



'.' attaches a class (type)

socrates:Human reads 'socrates is a Human'. The same colon is used for ?X:Mortal.



'<-' reads as 'if'

The rule's left side (the head) holds whenever the right side (the body) can be proven.



'?- ' starts a question

Asks the engine to prove the statement that follows, returning true or false (or unknown).

Concepts

World assumption

`world open.`

Declares how missing information is treated (open vs. closed world).

Fact

`socrates:Human.`

States that the individual socrates belongs to the class Human.

Rule

`?X:Mortal <- ?X:Human.`

If ?X is Human, then ?X is Mortal. '<-' reads as "if"; ?X is a variable.

Question

`?- socrates:Mortal.`

Asks the engine to prove whether socrates is Mortal.

```
quick-start.fx

world open.

socrates:Human.

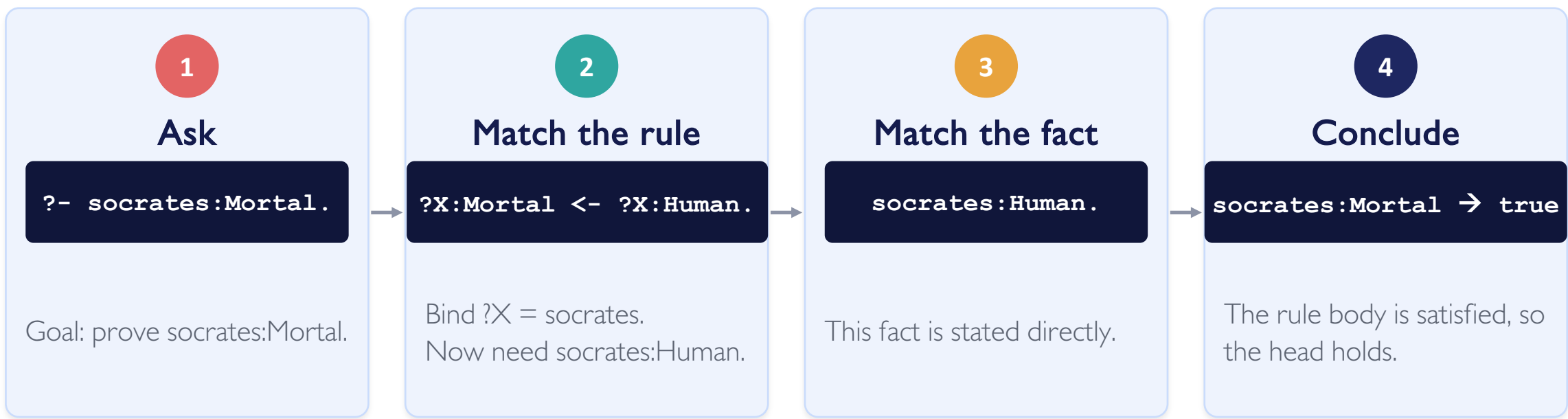
?X:Mortal <- ?X:Human.

?- socrates:Mortal.
```

Every statement ends in a period, the parser splits on it.

Evaluation

The question triggers a search. The engine tries to prove `socrates:Mortal` using every fact and rule it has.



Result: true, since the rule's condition (Human) was already proven by a fact.

Hide files Socrates.fx Saved Save Run program

```
1 world open.
2 socrates:Human.
3 ?X:Mortal <- ?X:Human.
4 ?- socrates:Mortal.
5
```

▼ Result completed Compact

true

Executed: Socrates.fx · saved version at run time

Logical truth (true, false, unknown) is separate from execution status.

► Execution details

FrameX

First Steps

FrameX Login

FrameX-Workbench

<https://framex.nlp-lab.ai/>



FrameX-Doc

<https://unisg-ics-dsnlp.github.io/FrameX-Doc/quick-start.html>

FrameX-CLI ... (more to follow)

[X]

FrameX-Workbench

NLP
LAB

HybridAI

Write F-Logic. Connect language and knowledge.
Build an agent that can explain its next step.



[X]

FrameX-Workbench

Workspace

Account

lorena.raichle@unisg.ch · student

Sign out

Your lab

running

Stop workspace

FILES

+ New

Hide files

classes-inheritance.fx

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Save

Run program

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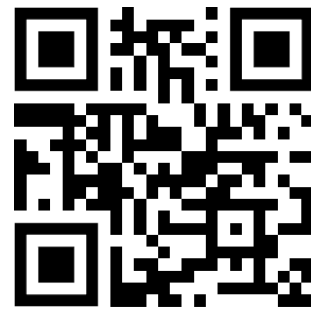
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► Execution details

FrameX

FrameX-Workbench

<https://framex.nlp-lab.ai/>



Initial Password:
userhsg2026

- 01 Open FrameX-Workbench
- 02 Login as **userXX** with initial password
- 03 Register via E-Mail, set your own password
- 04 Start your *Workspace*
- 05 Follow Quick Start Tutorial (until 4.6) + *Try it* Tasks

<https://unisg-ics-dsnlp.github.io/FrameX-Doc/quick-start.html>



[X] FrameX-Workbench



HybridAI

Write F-Logic. Connect language and knowledge.
Build an agent that can explain its next step.

Welcome to the lab

First visit: use your initial userNN account. After setup: use your registered email and personal password. Staff use their TA or teacher username.

Email or initial account

Password



Sign in →

► [Forgot password or cannot sign in?](#)



Götz-Henrik Wiegand



Lorena Raichle

Questions?

Contact us via hybrid-ai@mailman.nlp-lab.ai

Hackathon I

Derive from Knowledge

29.09.2026
14-18 Uhr